

Fertilization & Development

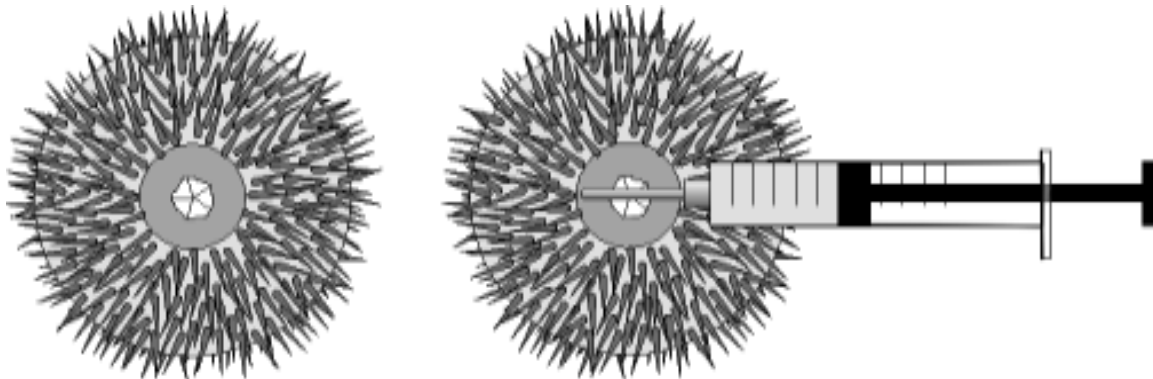
Today's Materials

- Live sea urchins from Niles Biological (4-5 per class, stored in fridge)
- 250 ml 1% acetic acid in seawater
- 25 ml 3 g commercial spermicide per 10 ml seawater
- 2L seawater stock
- 250 ml distilled water
- 1 small bottle of 0.5 M KCl
- Syringe with needles (for injecting urchins)
- Test tubes
- 12 150 ml beakers
- 12 10 ml graduate cylinders
- ~16 depression slides (will be washed/reused)
- Coverslips
- Disposable plastic pipettes
- Some ice buckets with ice
- 45 degree water bath
- Scissors and tape
- Blank Pages in Lab Notebook
- Extra Textbooks/Manuals in Room

Activity 1: Fertilization Using Live Sperm & Eggs

Fertilization in Sea Urchins and Humans

At the beginning of lab, your instructor will guide you in harvesting sperm and eggs from sea urchins by injecting them with a solution of KCl. This causes the urchins to release their gametes (sperm and eggs), which allows us to determine the sex of the urchins (either white sperm or yellow eggs will be released).



Experimental Procedures

Part A. Observing Fertilization Under Normal Conditions

1. EGGS SHOULD BE KEPT AT ROOM TEMPERATURE AND SPERM SHOULD BE KEPT ON ICE
 - a. Sperm are much more delicate than the eggs, so they must be kept undiluted on ice (the seminal fluid has substances in it to keep the sperm inactive and the ice also helps)
 - b. We will dilute the sperm in seawater to activate them prior to use on our slides.
2. We will use special “depression slides” to view urchin gametes. These slides have a small cavity in them that allows large cells, like eggs, more room than a flat slide.
3. **View eggs alone** by preparing a slide with a drop of eggs and a cover slip.
 - a. Be sure that the iris/diaphragm on your microscope (dial located under the stage) is closed so that only very little light is filtered through. The eggs should be relatively large and brownish in color. You should be able to see them easily under low power.
4. Next, get a graduated cylinder or test tube. Place ~2 ml of seawater in the tube and add two *drops* of concentrated sperm.
 - a. Keep this diluted sperm mixture on ice when not in use.
 - b. The concentrated sperm from the urchin is far too concentrated to view easily, so we always use diluted sperm for the activities today.
5. **View sperm alone** by preparing a slide with a drop of *diluted* sperm and a cover slip.
 - a. Again, be sure that the iris/diaphragm on your microscope is closed so only very little light is filtered through. Sperm are very tiny and almost clear, so good contrast is essential. You may need to work your way up to the “40x” objective lens power to see the sperm, though you should see their movement at “10x” power. You should look for *very* tiny specks jiggling around the slide.
 - b. *You will need to be able to observe sperm as part of your experiments that follow*, so it is important that you practice viewing sperm! ***Please don't hesitate to ask your instructor for assistance with viewing sperm!***
6. **View sperm motility and fertilization under normal conditions (positive control)**
 - a. Prepare a slide with a drop of eggs only.
 - b. Add one drop *diluted* sperm and place the cover slip on the slide.
 - c. Focus the microscope so you can see both the eggs and the movement of sperm.
7. Start timing and after 1 minute **record the percentage of fertilized eggs and amount of sperm movement** as described below.
 - a. A “fertilization membrane” forms immediately after fertilization to prevent polyspermy (fertilization by more than one sperm). The fertilized egg then begins to divide by mitosis. Eggs with fertilization membranes and eggs that are dividing should both be counted as “fertilized.”



Unfertilized egg



Fertilized egg with fertilization membrane



Fertilized egg that has started dividing

- b. At 1 minute, scan across your slide and look at 25 eggs. Record the number of out of those 25 that are fertilized. Multiply by 4 to get the percentage of fertilized eggs.
 - c. At the same time, briefly note down the level of sperm activity on a scale of 0-10 where 10 represents a high level of activity by all sperm cells and 0 represents no sperm activity whatsoever.
 8. Repeat this procedure to record the percentage of fertilized eggs and level of sperm motility after 5 minutes and 10 minutes in the tables on the designated page in your manual.
 9. **Perform a negative control:** *While your fertilization under normal conditions experiment is ongoing*, other members of your group should set up an experiment to observe fertilization with a negative control. We'll use **acetic acid** as the negative control. In humans, acetic acid is known to kill sperm and stop fertilization.
 - a. Mix 1 ml of diluted acetic acid with 1 ml of diluted sperm in a small beaker
 - b. Set up a slide of only eggs.
 - c. Place slide on microscope stage.
 - d. Add 1 drop of sperm diluted in acetic acid.
 - e. Start timing and record the percentage of fertilized eggs and sperm motility after 1 minute, 5 minutes and 10 minutes in the tables.
 10. Place used cover slips in the broken glass disposal bin. Wash depression slides for reuse in the next activity.

Part B. What Conditions Affect Fertilization Rates?

You will now test the ways different chemicals or temperatures affect fertilization. *Note that these conditions might not all inhibit fertilization. It's possible some could speed up fertilization!* This information is very relevant in a clinical sense where determining the reasons for infertility and finding treatments are important. You will select two conditions to test. Possible conditions include:

- ★ **Temperature** – low vs. high (incubating sperm in water baths)
 - ★ **Salinity** - higher and lower than sea water (ex. freshwater)
 - ★ **Detergents**
 - ★ **pH** – acid vs. base environment
 - ★ **Spermicide** – the most common component in spermicides is **nonoxynol-9**. Test what effect nonoxynol-9 has on sea urchin fertilization.
 - ★ **Other Chemicals** – Our lab room includes a number of chemicals you could mix with your samples to test their impacts. For example, alcohol, soap, disinfectant, etc. You might even have chemicals with you in your bag that you could test! (a drink/coffee)
1. Spend a few minutes designing your experiments.
 2. Check with your experimental design with your instructor before beginning.
 3. Carry out your experiment and record all your data in the tables below.
 4. ***Note that you have more than one microscope available at your table, so you can run the experiments at the same time!***
 5. Use the graph paper to create a point or line graph showing the amount of sperm motility at each time for each condition. You should create 1 graph with 4 sets of points or lines on it (ex. one line for the sperm motility information from each of your data tables).
 6. *Wash, dry and return* the depression slides you have used. Discard cover slips in the broken glass disposal bin.

Results/ Data Analysis**Positive Control** (normal fertilization)

	1 minute	5 minutes	10 minutes
# Fertilized eggs out of 25			
Percent fertilized eggs			
Sperm Motility (10=all cells highly active, 0=no activity at all)			

Negative Control (acetic acid)

	1 minute	5 minutes	10 minutes
# Fertilized eggs out of 25			
Percent fertilized eggs			
Sperm Motility (10=all cells highly active, 0=no activity at all)			

Condition #1 ()

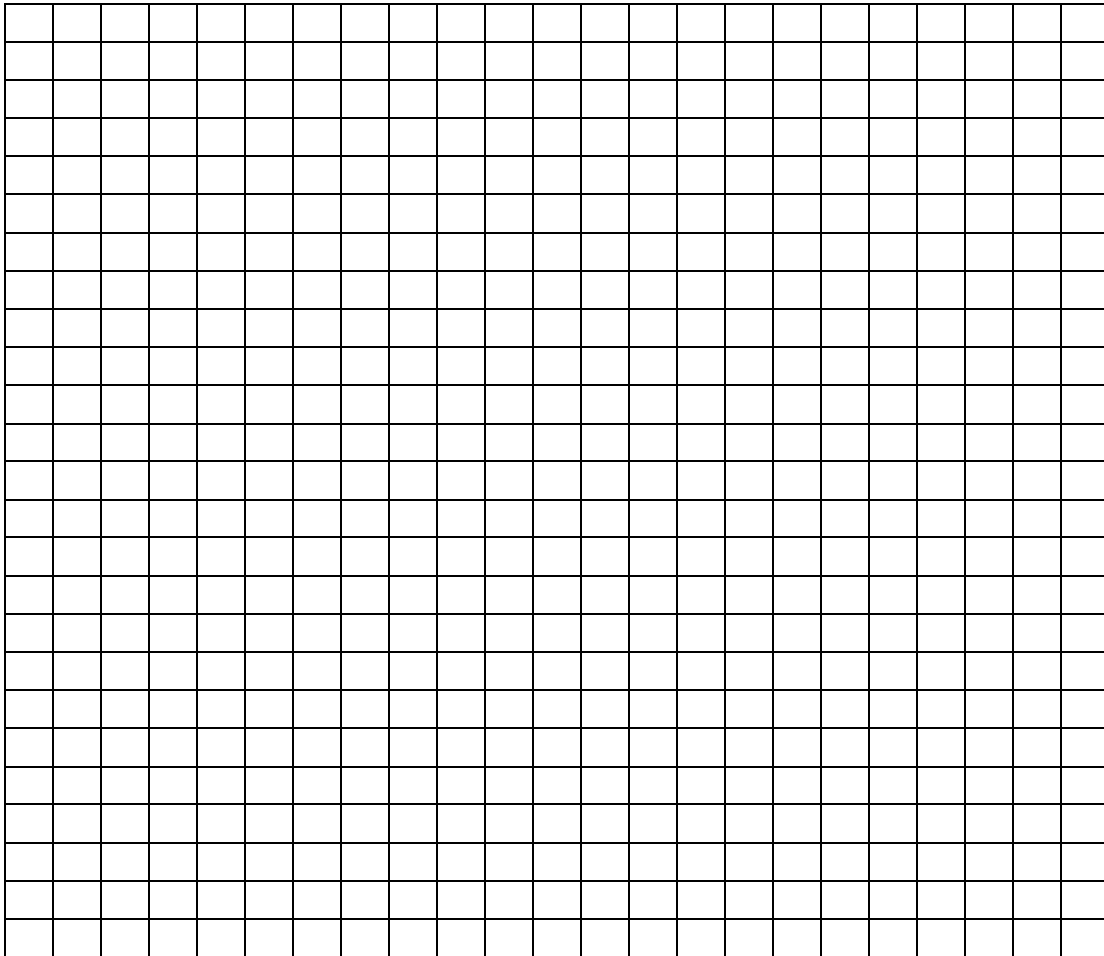
	1 minute	5 minutes	10 minutes
# Fertilized eggs out of 25			
Percent fertilized eggs			
Sperm Motility (10=all cells highly active, 0=no activity at all)			

Condition #2 ()

	1 minute	5 minutes	10 minutes
# Fertilized eggs out of 25			
Percent fertilized eggs			
Sperm Motility (10=all cells highly active, 0=no activity at all)			

III. Create a **point** or **line** graph for the sperm motility data from Part II and Part III. You should make one graph with four sets of points or lines on it (ex. one line representing the motility information from each of your data tables). Make sure your graph is clear, has scales, labels, and a key/legend to show what data is represented by each set of points.

Sperm Motility Under Different Conditions



1. Spermicides are substances used to try to kill sperm and prevent fertilization. While spermicides alone are generally not as effective as other types of birth control, they are often used in combination with other products like condoms. Given your results, make an argument to a drug company for a new type of spermicide using one of the chemicals you tested. Assuming it was safe, which chemical would you propose for a new spermicide? Describe the experimental evidence you would present to the company to back up your proposal.
2. Perform some research on the “cortical reaction” (i.e., the formation of a fertilization membrane) which is what you studied in this activity! Based on what you read, create diagrams below to summarize the main steps in the cortical reaction. Make sure to show the roles of the zona pellucida, the fertilizing sperm cell, calcium, proteases, and peroxidases.



Activity 2. Homologues in the Human Reproductive System

Even though males and females appear to have quite different reproductive parts, many of those parts are very similar in structure and function between males and females. Body parts that come from the same tissues during early development are called “homologues.” Cut out the squares on the following page, and use the information on the squares to try to determine which parts are homologous between males and females (derived from the same tissues). Tape those parts next to each other in the spaces below, and explain your reasons for pairing up those parts.

Male Reproductive Parts**Homologous Female
Reproductive Parts****Why do you
think these parts
are *homologous*?**

Cut out these squares along the dotted lines, then follow the instructions on the previous page.

	Secretes a fluid to flush out and lubricate the urethra in preparation for sperm travel.		Covers and protects deeper reproductive organs
	Erectile organ for intercourse. Sensors in skin provide sensations of pleasure.		Site of meiosis to produce eggs. Hormone production.
	Secretes a substance that helps form semen, which is ejaculated along with sperm cells.		Fills with blood during sexual stimulation. Involved in feelings of arousal
	Protect the testes and maintain appropriate temperatures for developing sperm		Secretes fluids that might be related to female ejaculation
	Produce male sex hormones. Form sperm by meiosis.		Secretes mucus that serves as lubrication
Testes		Skene's Gland (near urethra)	Bartholin's Gland (posterior to vagina)

1. An unusual genetic condition runs in Jorrel's family that causes the prostate gland to develop slowly and incompletely. Jorrel is worried about passing this condition on to his children, since his wife is pregnant. However, Jorrel's baby will be a girl and will therefore not have a prostate. Given the work you did in Activity 2, what part in the female reproductive system do you predict might be impacted by this condition?

What symptoms might a female experience if that part of the body does not fully develop?

Activity 3: Development After Fertilization

One of the striking features of reproduction and development is that we all start as just a single cell – a fertilized egg. All of the amazingly diverse types of cells now in your body are derived from that one original cell. As that fertilized egg divided, some of the resulting cells began to specialize. Then some of those cells further specialized and some of those further specialized, etc., until you had a wide variety of highly specialized and unique cells. In this activity, we will explore the origins and lineages of various specialized cells in your body.

1. Each person at your table will be assigned a specialized, mature cell type. Your job is to figure out where your cell came from, in terms of its lineage back to an embryonic stem cell.
2. Perform some image searches (e.g., images.google.com) where you search for your cell type along with words like Stem Cell Lineage or Stem Cell Differentiation. Scroll through the results looking for images that look like family trees or flow charts (circles or pictures with branching arrows). Consult multiple pictures if necessary.
3. Create a list to describe how your cell is created, starting from an embryonic stem cell. For example, below is a sample for how an adipocyte is created:

Embryonic Stem Cell → Mesoderm Cell → Mesenchymal Stem Cell → Pre-Adipocyte → Adipocyte

Make your similar list below with your cell as the end point.

4. We will put some of this together as a whole class and try to create a diagram representing all of the cell relationships we researched. Record that diagram below and be sure to note any surprising relationships between cells that are discussed!